

IMPORTANT

1. DO NOT FORGET TO RECORD YOUR SCREEN
2. USE OF AI TOOLS OR ANY OTHER AIDS NOT ALLOWED

Grade 3:

Use the entry ticket as the base and make the following changes:

- Create a counter variable that is initially assigned the integer value 0.
- Extend what happens when the user adds a new item to the shopping list so that it does not only add the new item to the string, but it also increases the counter variable by one.
- Add a new menu option called **Show number of added items** that prints how many items have currently been added to the shopping list. The user shall be able to pick the new menu option using the letter **S** when selecting a menu option.
- Use the following format when printing the number of items on the list:

Currently the list contains 3 item(s)

NOTE: The formatting of the final shopping list shall remain the same, it shall not print the number of items added. One shall only be able to see the number of added items if one uses the new menu option.

Grade 4:

After passing all the automated tests for grade 3, continue by adding the following functionality:

- Change how the menu option **Add an item** works by introducing a restriction on how many items the shopping list can contain.
- The limit for how many items the user is allowed to add to the shopping list shall be 10, and if they try to add more than 10 items they shall be met with the following error message:

ERROR: The list is full, your item was not added

- After the error message has been displayed the user shall be returned to the menu where they are free to make another choice.

NOTE: When the user is returned to the menu after having received an error message there is nothing stopping the them from trying to add another item, but if they try, they shall be met with the same error message as before and then be returned to the menu again.

Grade 5:

After passing all the automated tests for Grade 4, continue by adding the following functionality:

- Update the message shown when selecting the menu option **Show number of added items** to also include a graphical visualization of how full the shopping list is.
- The graphical visualization shall be printed on its own line, immediately following the line with information about how many items have currently been added to the shopping list.
- The graphical visualization shall have the following format where the **start and end** is represented by square brackets, the number of **items on the list** is represented by stars, and the number of empty **slots left** on the shopping list is represented by spaces.

```
[****      ]
```

In the example above we can see that the user has added **4 items**, and that they **can add 6 more** items if they would like to. There space between the square brackets shall always contain ten characters, stars and spaces combined.